AI-Based Anomaly Detection and Intelligent Monitoring of Chiller Operating Data

Case study 11 - Impact of condenser temperature and load on efficiency

Track: Explainability, benchmarking and robustness

Method: LOF + local feature perturbation

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Summary

This report asks if local explanations can help to understand why individual chiller log records have high anomaly scores in a Local Outlier Factor (LOF) model. The instructor's log has 522 hourly records for five chillers, covering 10 days. After rule-based data cleaning, train-only imputation, scaling and chronological split, we fit an LOF model (k=25) on the first five days. We explain high-scoring records by replacing one feature (or one group of subsystem features) at a time with values from the nearest training neighbours. We measure how much the score drops.

The explanations are reproducible: 96% and 97% of the re-draws/noise tests have the same top feature, 86-97% of all stability tests have the same top subsystem. In a synthetic test with a separate copy of clean test data, the injected feature is top in 94% of detected cases. The case review reveals two things that are not visible with the score alone. Four chillers have a one-hour condenser water excursion to about 35°C at the same time - a plant-level effect. One record is likely a recording error. A flagged record is only a potential anomaly. None of these is confirmed equipment faults.

1. Research question and approach

Research question: Can local explanations help to understand why records have high anomaly scores? No fault labels exist in the log, so detection accuracy cannot be measured. Instead, we evaluate the explanations using the assigned methods of explanation stability and local case review. We also add two label-free checks for the adequacy and plausibility of the explanations. The case study (condenser temperature and load versus efficiency) provides the engineering context. It gives the operating plane in which we read scores.

2. Data and quality assessment

There are two title rows in the sheet, followed by 522 records x 60 columns. They are sorted by time (descending). Each record is one chiller, with two compressors. We create hourly timestamps by combining Date and Time. No duplicates exist. The chillers are cycled on/off, so each chiller has gaps (up to 113 h). The data is not a regular series for each chiller. So we do not use sequence models. There are no blank cells. But during the audit, we found the following problems.

Finding	Evidence	Decision
Hidden bad values	28 cells across 21 rows contained implausible values, such as 2196 A motor current, 631 °C liquid temperature, 40.8 V and 85 °C condenser outlet.	Values outside engineering limits were converted to missing values (NaN), recorded and imputed.

Finding	Evidence	Decision
Compressor off	Five rows had zero compressor load and zero values for all compressor variables.	Treated as valid operating conditions and excluded from LOF scoring.
Redundant columns	Load percentage was derived from motor current and full-load current (2.25 and 2.15 A per %). Lift, approach and superheat were derived from other measurements.	Redundant columns were removed, retaining 23 independent features.
Run-hour counters	Large jumps, such as 1179 and 40715, were observed against an approximate value of 11,500.	Audited but excluded from model features.
Pump pressure zeros	Condenser pump pressure was recorded as 0.0 in approximately 5% of rows despite normal pump current.	Excluded because the pressure gauge readings were not logged.
Mixed units	Temperatures were recorded in °C, compressor pressure in kg/cm² and pump pressure in bar.	Pressure variables were excluded from model inputs; therefore, unit conversion was unnecessary.

Table 1. Data quality findings and decision for each

Finding Evidence Decision

Hidden bad values: 28 cells in 21 rows, e.g. 2196 A motor current, 631°C liquid temperature, 40.8 V, 85°C condenser outlet. Outside engineering range -> NaN, logged, imputed
Compressor off for 5 rows with load = 0; all compressor variables = 0. Valid operating state. No LOF score.
Redundant columns: Load % = motor current / full load amps (2.25 and 2.15 A per %). Lift, approach and superheat columns are differences of others. Dropped. We keep 23 independent features.
Run hour counters jump like 1179 or 40715 vs ≈ 11 500 audited, but not used as features.
Pump pressure zeros: Condenser pump pressure is 0.0 on ≈ 5% of rows with normal current. Excluded (no gauge logged).

Mixed units: Temperature is °C. Compressor pressure is kg/cm². Pump pressure is bar. Pressures are not model inputs. No conversion required.

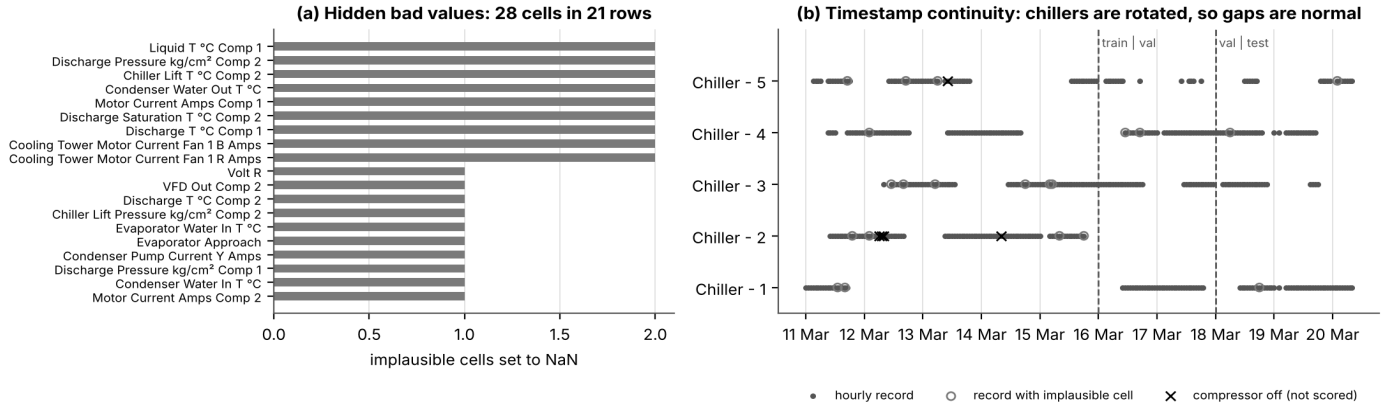


Figure 1. (a) Implausible cells by variable. (b) Record availability per chiller, with train/validation/test boundaries.

3. Preprocessing

All statistics are calculated on the training block only. Chronological split (whole days). Train 11-15 Mar (282 scored rows). Validation: 16- 17 Mar (115 rows). Test 18-20 Mar (120 rows).

• Imputation. Only 11 cells (0.09% of scored cells, 8 rows) need imputation. We fill a cell from the previous record for the same chiller if it is ≤ 3 h old. Otherwise, we fill it with the training median for this chiller. No interpolation across a gap. No future value is used.

• Features. There are 24 candidate features. They cover five subsystem groups: condenser side, load and capacity control, evaporator side, compressor/refrigerant circuit, electrical supply. We prune at training |Spearman| > 0.90. One feature is removed (EXV ΔP of compressor 2). We keep 23 features. We do not use PCA. This keeps explanations in physical units.

• Scaling. We use RobustScaler (median/IQR). Fitted on training rows. This is distance-based LOF. There are still extremes in the training block (up to 12 robust units).

• Neighbourhood. We use k to define "local". We refit LOF for each k in {5…60} on 40 random 80% subsamples of the training block. We record the Spearman correlation between validation scores and the full fit scores. We choose the smallest k that is within 97% of the best stability. It must be at least d+1 = 24. k=25, ρ=0.96 (Figure 2).

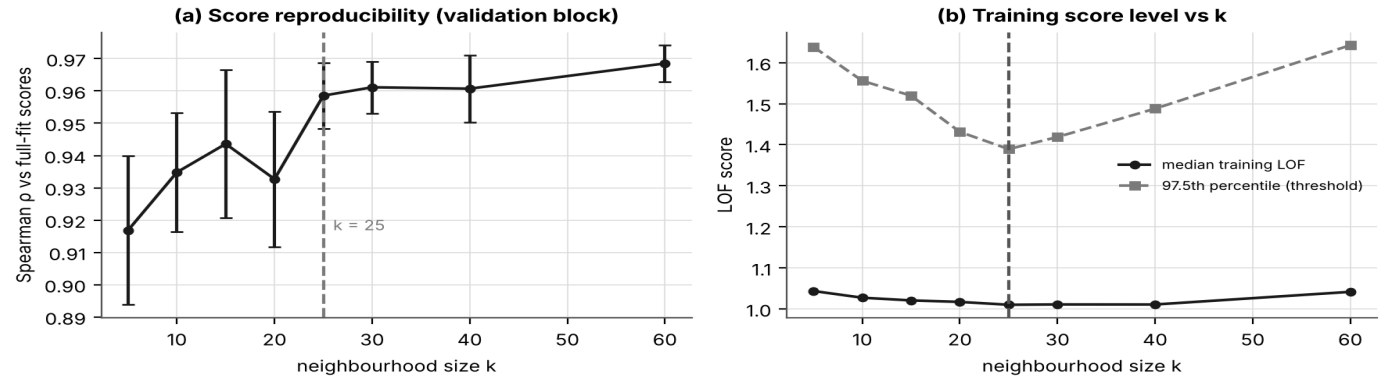


Figure 2. Choice of k. (a) Validation score rank stability. (b) Training score level.

4. Method

4.1 LOF anomaly score

LOF (scikit-learn, novelty mode, Euclidean distance, k=25) compares the local density around a record with that of its neighbours. A score around 1 is normal. Larger values indicate the record is locally sparser than its neighbours. The alert threshold is the 97.5th percentile of the training scores, i.e. 1.389. A flag is just "unusual compared with 11-15 March". No more. See model_config.json for details.

4.2 Local feature perturbation

For record x with score s(x). The explainer finds its 25 nearest training neighbours. It replaces feature j with the value of a randomly drawn neighbour (50 draws). It recomputes LOF. It reports Δj = s(x) - mean perturbed score. If Δj is large, the score would drop significantly if that feature looks like the neighbours. We do the same for whole subsystem groups. We draw one neighbour for the group. We preserve the within-group correlation (e.g. condensing temperature of both compressors). The replacement values are real values from the same operating neighbourhood. They are within the range of observation. They are consistent with the local state. We use a naive baseline for comparison. It replaces a feature with the global training median.

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5. Case study: condenser temperature, load and efficiency

The log has no power or flow readings. So we cannot compute kW/TR. We assess efficiency with two thermodynamic proxies. Compressor lift (condensing minus evaporating temperature) and ideal Carnot COP. We regress on condenser water inlet temperature and mean running load. We only use training rows for the fit. We check with 1000 bootstrap samples. We check on the later days.

Effect	Change in Lift (°C)	95% Confidence Interval	Change in Ideal COP	Held-out R^2 (Lift / COP)
+1 °C condenser water inlet temperature	+0.59	0.35 to 0.81	−1.8%	0.49 / 0.41
+10% mean running load	+0.45	0.31 to 0.63	−1.5%	0.49 / 0.41

Table 2. Impact of condenser water temperature and load (training fit, last column: R^2 on later days)

Effect Lift (°C) 95% CI Ideal COP Held-out R^2 (lift / COP)
+1°C condenser water inlet +0.59 0.35 to 0.81 -1.8% 0.49 / 0.41
+10% mean running load +0.45 0.31 to 0.63 -1.5% 0.49 / 0.41

Both a hotter condenser water supply and a higher load increase lift. This reduces the ideal COP. The two variables are also correlated (ρ=0.67). We only have 10 days, so the separate impacts are uncertain. They are proxies for efficiency. Not measured efficiency. Figure 3 shows the same plane, but with flagged records. The highest scores are in the hot condenser corner. This was not covered in the training days.

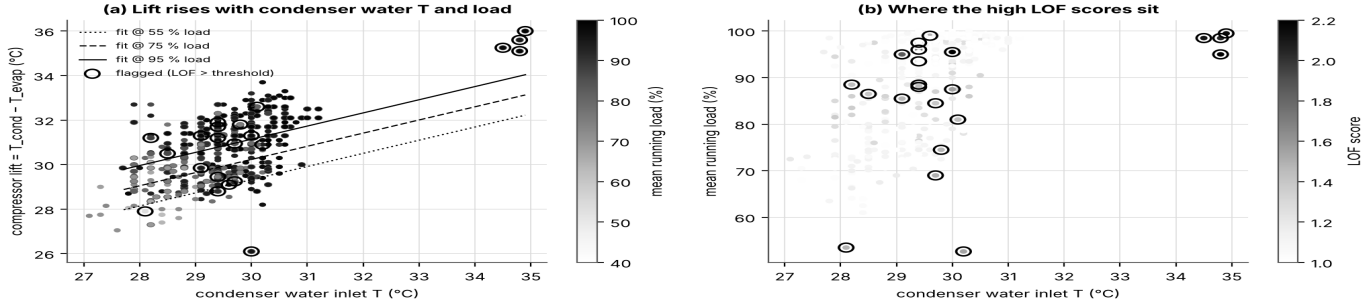


Figure 3. (a) Lift vs condenser water inlet temperature. Shaded by load. (b) Same plane. Shaded by LOF score. Circles are flagged records.

6. Results

6.1 Anomaly scores

The flag rate is 2.8% in training (8/282). 4.3% in validation (5/115). 7.5% in test (9/120). Most excess in the test is due to the cluster at 13:00 on 18 March. There are two other records (Figure 4). The median score is around 1.0 in all three blocks. So we score ordinary operation consistently over time. The highest score in training (2.53, Chiller 3, 14 Mar) is included in the training data. There is no label to confirm it is anomalous. It reminds us that the training block is not guaranteed to be clean.

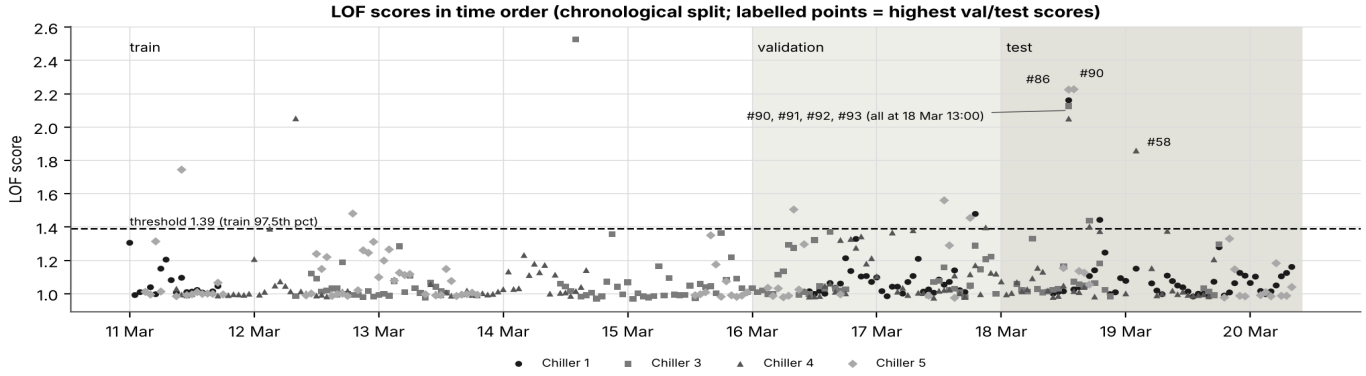


Figure 4. LOF scores in time order. Training threshold. Chronological blocks.

6.2 Local case review

We review the 15 highest validation/test scores (all 14 flagged + next). Figure 5 shows which subsystem group is responsible for each score. Figure 6 shows the leading features for four cases. We show the value observed next to the median of the 25 nearest training neighbours.

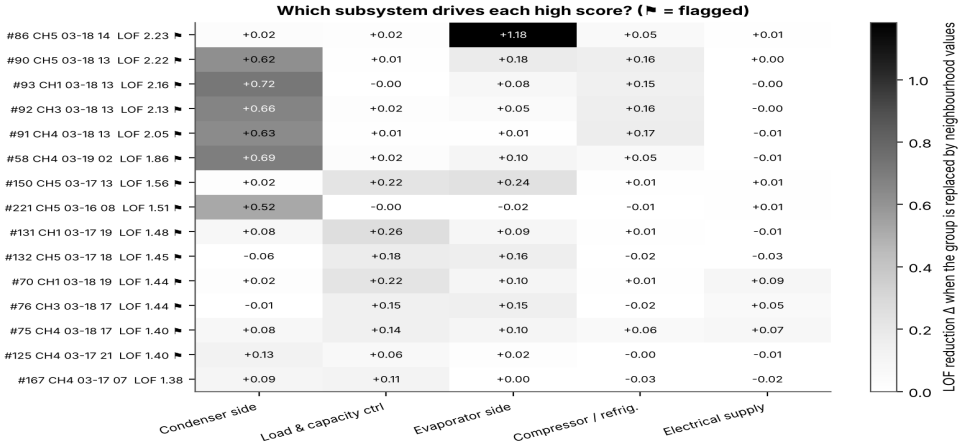
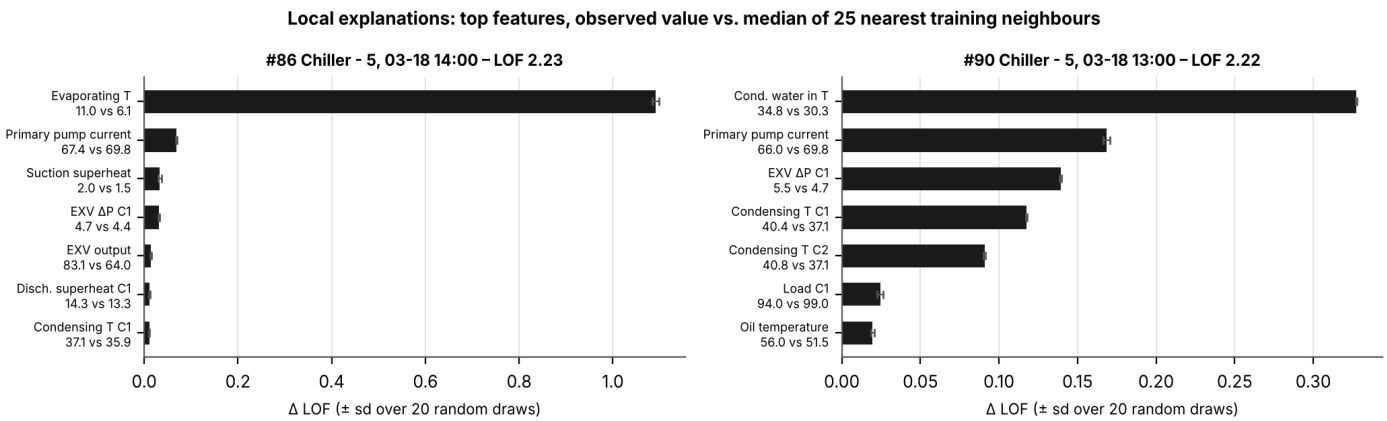


Figure 5. Subsystem-level attribution of the 15 highest scores (⚑ = flagged).



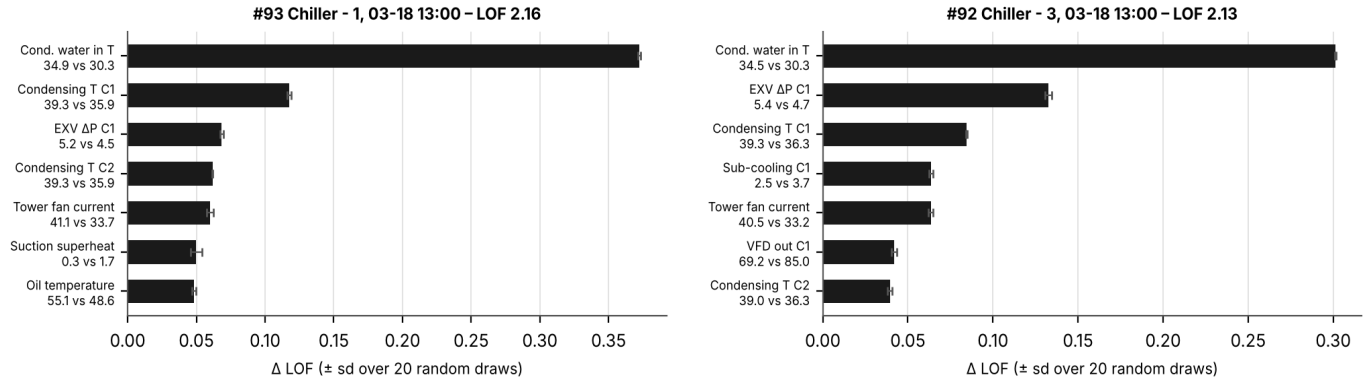


Figure 6. Feature-level explanations for the four highest scores. Bars ± SD over 20 draws.

Records	LOF Score	Main Explanation	Cross-Check and Interpretation
#90–#93 (CH5, CH1, CH3, CH4), 18 March, 13:00	2.05–2.22	Condenser-side variables (Δ = +0.62 to +0.72). Condenser water inlet temperature was 34.5–34.9 °C compared with 30.3 °C in neighbouring records.	The training-period maximum was 30.8 °C. Compared with the previous hour, condensing temperature increased by 3.4 °C, lift increased by 3.9 °C, ideal COP decreased by 11%, and cooling-tower fan current increased by 24%. The simultaneous change across four chillers suggests a possible plant-level event. The exact cause cannot be determined from the available data.
#86 (CH5), 18 March, 14:00	2.23	Evaporating temperature was 11.0 °C compared with 6.1 °C in neighbouring records (Δ = +1.09). Other feature contributions were below 0.07.	Evaporator approach and suction superheat were both inconsistent by approximately 6.0 K. The recorded evaporating temperature also exceeded the chilled-water outlet temperature of 8.3 °C. This suggests a probable recording error.
#58 (CH4), 19 March, 02:00; #221 (CH5), 16 March, 08:00	1.86; 1.51	Condenser-water temperature differences were 1.4 K and 0.4 K, compared with neighbouring values of 2.6 K and 2.2 K.	Possible causes include high condenser-water flow or sensor offset. Further inspection of flow measurements and maintenance records is required.

Records	LOF Score	Main Explanation	Cross-Check and Interpretation
#131, #132, #70, #76 and #75	1.40–1.48	Unequal compressor loads, such as 52% and 86%, with a recorded supply voltage of 404 V.	These observations appear consistent with compressor load sharing and sequencing. They are considered potentially benign operating variations.

Table 3. Summary of main cases. (potential anomalies, not confirmed faults)

Records LOF Explanation Cross-check and reading

#90-#93 (CH5, CH1, CH3, CH4) 18 Mar 13:00 2.05-2.22 Condenser side (Δ +0.62 to +0.72) Condenser water in 34.5-34.9°C vs 30.3°C neighbours. Training maximum was 30.8°C. vs an hour before. Condensing T +3.4°C. Lift +3.9°C. Ideal COP -11%. Tower fan current +24%. All four chillers at the same time. Plant-level cause. Not clear from the log.

#86 (CH5) 18 Mar 14:00 2.23 Evaporating T 11.0°C vs 6.1°C (Δ +1.09. All others < 0.07). Logged evaporator approach and suction superheat off by 6.0 K. 11.0°C > chilled water outlet 8.3°C. Probable recording error.

#58 (CH4) 19 Mar 02:00. #221 (CH5) 16 Mar 08:00 1.86. 1.51 Condenser water ΔT: 1.4 K, 0.4 K vs 2.6 K, 2.2 K. Possible high condenser water flow. Sensor offset. Need to check flow or maintenance.

#131, #132, #70, #76, #75 1.40-1.48 Unequal compressor loads (e.g. 52% vs 86%). Supply voltage 404 V. Load sharing and sequencing consistent. Probably benign.

The case review shows what we learn from the explanations. Without explanations, #86 and #93 are both "high score". With explanations, one is a suspicious reading that fails an independent physical identity check (we did not use the identity in training). The other is about the condenser water temperature. This is consistent with the mechanism in the case study. A hotter condenser increases lift. There are no imputed cells in the top 15. So the scores are not artefacts of imputation.

7. Evaluation of explanations

7.1 Stability

We compare the base explanation with four types of variation for the 15 cases. Other random draws. Other neighbourhood sizes (k=10, 15, 40, 60). 20 refits on a random 85% of the training data. (refit scaler and LOF). We add Gaussian noise of 0.05 robust units to the record. (approx. logger resolution 0.1 units).

Variation	Spearman ρ (Features)	Top-1 Feature Agreement	Top-3 Jaccard Index	Top-1 Group Agreement
Repeated random draws	0.95	96%	0.91	97%
Measurement noise	0.90	97%	0.89	96%

Variation	Spearman ρ (Features)	Top-1 Feature Agreement	Top-3 Jaccard Index	Top-1 Group Agreement
85% training data resampling	0.78	89%	0.76	86%
Different neighbourhood sizes (k)	0.75	86%	0.77	89%

Table 4. Explanation stability. (mean over 15 cases. Agreement with base explanation)
Variation Spearman ρ (features) Top 1 feature same Top 3 Jaccard Top 1 group same
Other random draws 0.95 96% 0.91 97%
Measurement noise 0.90 97% 0.89 96%
85% training resample 0.78 89% 0.76 86%
Neighbourhood size k 0.75 86% 0.77 89%

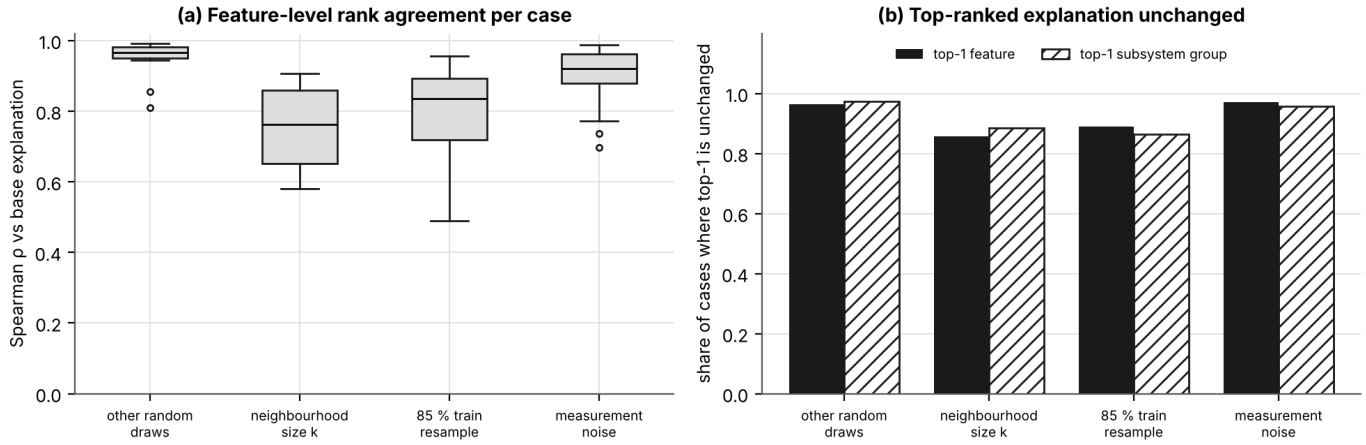


Figure 7. (a) Per-case rank agreement. (b) Share of cases with unchanged top-ranked feature/group. The dominant reason is reliable. But the order of weak contributors is not. The agreement drops to ρ≈0.75 when we change k or the training sample. The subsystem-level answer is more robust than the single-feature ranking. So we should start with the group in reports. We use the feature ranking as supporting details.

7.2 Sufficiency, plausibility and injection test

• Sufficiency. If we replace the top-ranked group with neighbourhood values. 10/14 flagged records (71%) are below the threshold. 21% for the random group. We remove the median 52% of the excess score.

• Plausibility. If we perturb normal records feature by feature. Only 0.4% look anomalous with neighbourhood replacement. 1.2% for global median. 2.4% for random training value. 4.3% for uniform draw from training range.

• Injection test. We take a copy of 60 clean test records. We shift one feature at a time by 2-6 robust units. We clip to the training 0.5-99.5 percentile range. We do not touch the clean reference data.

Single feature shifts are often diluted in 23 features. So LOF only flags 366/5139 injections (3%, 11%, 15% for 3, 4, 6 unit shifts; essentially none for 2 units).

Explainer	Top-1 Feature	Top-3 Features	Top-1, All Injections
Local perturbation (proposed method)	94.0%	98.1%	70.8%
Naive median replacement	92.1%	95.6%	64.0%
Largest |z| (without LOF)	81.1%	94.8%	52.7%
Random ranking	4.4%	12.8%	4.3%

Table 5. Share of flagged injections where the injected feature is top-ranked / top 3. (95% CI for top 1 about ±2.4 to ±4 points)

Explainer Top 1 feature Top 3 features Top 1 all injections

Local perturbation (this work) 94.0% 98.1% 70.8%

Naive median replacement 92.1% 95.6% 64.0%

Largest |z| (no LOF) 81.1% 94.8% 52.7%

Random ranking 4.4% 12.8% 4.3%

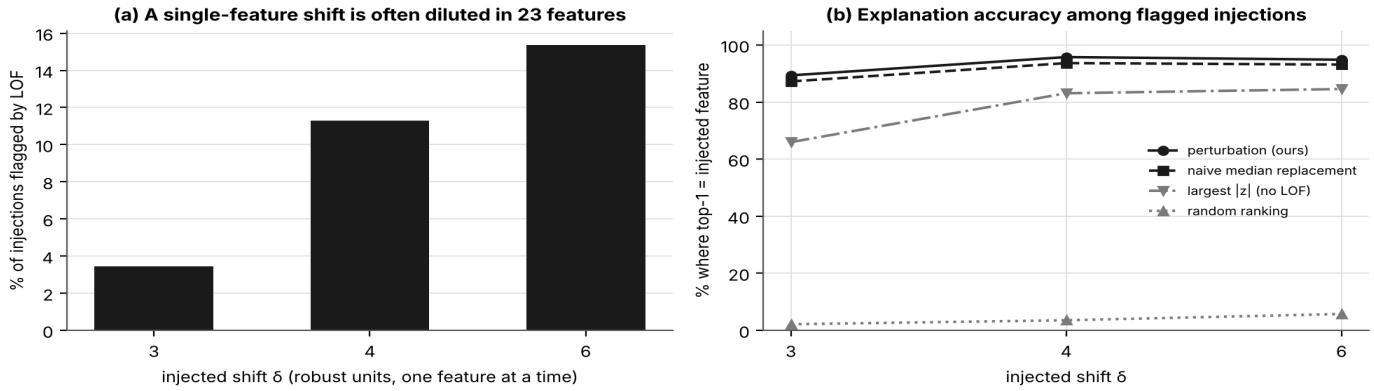


Figure 8. (a) Share of injections flagged by LOF. (b) Top-1 accuracy of each explainer among flagged injections.

The local perturbation explainer is clearly better than plain largest deviation ranking. It is far above chance. It is slightly better than naive median replacement (2 points for flagged cases; 7 points for all injections). But the injected shifts are the easy case (single feature change). It shows the explainer is sound. Not that it will find the cause of every real fault.

8. Discussion and limitations

Yes, within limits. Local explanations can identify the dominant feature or subsystem behind a high score reliably. They express it plainly ("34.8°C vs 30.3°C in similar states"). They separate a probable recording error from a plant-level event. They are connected to the case study mechanism. But they do not establish a cause. The ranking of minor contributors changes with k and the training sample. Every finding in Table 3 needs independent evidence. E.g. trend data, maintenance records, sensor checks.

• No power or flow data. So efficiency is a proxy. No fault labels. So no precision/recall. Only 10 days. Only 282 training rows for 23 dimensions.

• Hourly records of the same chiller are near duplicates. The chillers are pooled. So we would need much more data per chiller model. Threshold based on the training block. There may be unusual states in the training block (2.53 score).

• We choose k only based on validation block stability. The injection study only covers single feature shifts. No test for multi-feature drifts. E.g. slowly fouling condenser.

9. Conclusion

We have stable, readable, mostly sufficient explanations of why chiller records are scored as unusual. This is based on LOF with neighbourhood-based feature perturbation. The strongest finding is a one-hour, four-chiller condenser water excursion to about 35°C. This was not covered in training. We estimate an 11% loss of ideal COP. The second finding is a single reading that fails a physical consistency check. Both are potential anomalies. We need to investigate with independent evidence.

Reproducibility

Files: chiller_lof_explainability.ipynb (executed, seed 42). preprocessing_notes.md. model_config.json. metrics.json. tables/*.csv. figures/*.png. Python 3.13. scikit-learn 1.9. pandas 3.0.

References

Each reference is followed by the sections of this report where it was used.

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[2] Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830. *(Used in: Sections 3, 4.1 (LocalOutlierFactor, RobustScaler))*

[3] Chandola, V., Banerjee, A. and Kumar, V. (2009). Anomaly detection: A survey. *ACM Computing Surveys*, 41(3), Article 15. *(Used in: Sections 1, 8 (unlabelled anomaly detection))*

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[5] Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. *(Used in: Section 4.2 (feature attribution by replacing features))*

[6] Hooker, S., Erhan, D., Kindermans, P.-J. and Kim, B. (2019). A benchmark for interpretability methods in deep neural networks. *Advances in Neural Information Processing Systems*, 32. *(Used in: Section 7.2 (sufficiency and plausibility of replacement))*

[7] Doshi-Velez, F. and Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv:1702.08608*. *(Used in: Section 7 (evaluating explanations without labels))*

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[11] Course dataset: *Analyse_Chillers_AI_Course.xls* (sheet Log_Chiller_South), hourly chiller operating log, 11-20 March 2026, supplied by the course instructor. *(Used in: Sections 2-3 (data))*